ISSUES

Policy Brief #3
September 2025

Intimate Partner Violence & Brain Injury in U.S. Adults: Findings from the Medical Expenditure Panel Survey, 2021

National estimates show that adults who screen positive for intimate partner violence (IPV) have disproportionately higher rates of injury-linked emergency and inpatient encounters and substantially higher per-capita head/brain-injury–linked medical spending than adults who screen negative.

Key messages

- Approximately 9 million adults screened positive for IPV in 2021 in the MEPS data.
- Respondents with a positive screen had nearly three times higher ER rates and about two times higher inpatient rates than screen-negative adults (Table 1).
- Brain-injury events rise from 1.25 to 3.13 per 100 people, and per-capita brain-injury costs nearly double in the IPV-positive group (Table 7).
- Inpatient traumatic brain injury (TBI) events are less frequent than ER TBI events but have higher spending per discharge (Tables 5 and 6). IPV-positive adults carry a disproportionate burden of these injuries overall (Table 7).
- Poverty, social stressors, depression, and chronic conditions are significantly associated with IPV screen positivity in adjusted models (Tables 2–4).

Background and context

IPV is a known cause of traumatic injury, including TBI.[1–3] Using nationally representative data from the 2021 Medical Expenditure Panel Survey Household Component (MEPS-HC), we examined associations between IPV, measured using a previously validated screening tool,[4,5] and injury patterns and costs, focusing on TBI. Understanding how a high-risk group of adults who experience intimate partner violence (IPV-positive adults) disproportionately contributes to acute injury care can help inform targeted screening, referral, and care coordination strategies across emergency, trauma, and behavioral health services systems.

Details of the research

Overview of methods: We identify intimate partner violence (IPV) screen-positive adults using the Hurt, Insult, Threaten, Scream (HITS) scale (HITS ≥10 = positive; HITS <10 = negative). The HITS scale is comprised of four questions: "How often does your partner: (1) physically hurt you, (2) insult you or talk down to you, (3) threaten you with harm, and (4) scream or curse at you?" Patients responded to each of these items with a 5-point frequency format: never, rarely, sometimes, fairly often, and frequently (Range = 4 – 20). We coded injuries using ICD-10-CM and CCSR codes from the Conditions data and linked the data to event files for type of care (ER, inpatient, outpatient, office-based, home health) and prescriptions, through the MEPS condition–event linkage. Brain injury is defined as TBI (S06) and other head injuries (S00–S09). Outcomes include injury events per 100 adults, per-capita injury spending, and brain-injury–specific counts, rates, and costs. Adjusted comparisons use survey-weighted GLMs (*svyglm*) with standardization over age, sex, and race/ethnicity.

Key finding: About 9 million adults (3.6%) in the U.S. screen IPV-positive, but this group carries a disproportionate burden of injury, including head/brain injury, and markedly higher ER/inpatient use and spending. IPV positivity is linked to a greater than two-fold increase in injury and brain-injury burden and brain-injury costs per person, and a three to fourfold increase in acute-care use.

Summary of the research questions and objectives.

- RQ1: What is the national prevalence of IPV screen positivity and the associated burden of injury and brain injury?
- RQ2: How large is the brain-injury burden among IPV-positive adults (i.e., number, events per 100, and per-capita dollars) compared to adults who did not screen positive for IPV?
- RQ3: What sociodemographic, social, and medical factors are associated with a positive IPV screening (adjusted prevalence and risk ratios (RR)?

RQ1: National burden of IPV positivity and associated injury?

- **Prevalence**: 3.6% of adults are IPV-positive (mean 0.0358; SE 0.002). This represents approximately ***9 million adults with a positive IPV-screen nationally***.
- **Injury utilization** (adjusted, per person unless noted) (Figure 1):
    - IPV-positive adults have 9.17 emergency visits per 100 vs 3.22 among IPV-negative adults, a 185% increase..
    - IPV-positive adults have 3.13 inpatient stays per 100 vs 1.57 among IPV-negative adults, a 99% increase.
    - Per-person ER spending averages $110 for IPV-positive vs $40.1 for IPV-negative, a 174% increase.[1]
    - Per-person inpatient spending averages $382 for IPV-positive vs $191 for IPV-negative, a 100% increase.
    - Per-person prescription spending is $17.2 for IPV-positive vs $23.3 for IPV-negative, a 26% *decrease*.

RQ2: Magnitude and distribution of brain injury by IPV screen?

- *Brain-injury prevalence*: Brain injury prevalence is 1.7 times higher among individuals with a positive IPV screen (0.75/0.44 ≈ 1.70).
- *Brain-injury events per 100 adults*: IPV-positive adults have 3.13 events per 100 vs. 1.25–1.97 among those who screened negative, representing almost twice as many events.
- *Per-capita brain-injury dollars*: Adults who screened positive for IPV spent twice as much per person on brain-injury–linked care than those who screened negative (≈$39.6 vs $16–$22), averaged across all adults.[2]

TBI/head-injury–linked spending among adults who screen positive for IPV. Per-capita head/brain-injury–linked spending (Emergency Room (ER), Inpatient (IP), Outpatient (OP), Office Based (OB), Home Health (HH), Prescriptions (RX); excluding dental) is about 2 times higher among adults who screen IPV-positive than among those who screen negative (≈$39.6 vs $16–$22; survey-weighted; all adults included, $0 for non-users).

RQ3: Demographic and social context (adjusted prevalence & RRs)?

Who screens positive? Per-adult (population-averaged) spending on head/brain-injury–linked care (ER, IP, OP, OB, HH, RX; excluding dental) is about two times higher among IPV-positive vs IPV-negative adults (≈$39.6 vs $16–$22).[i]

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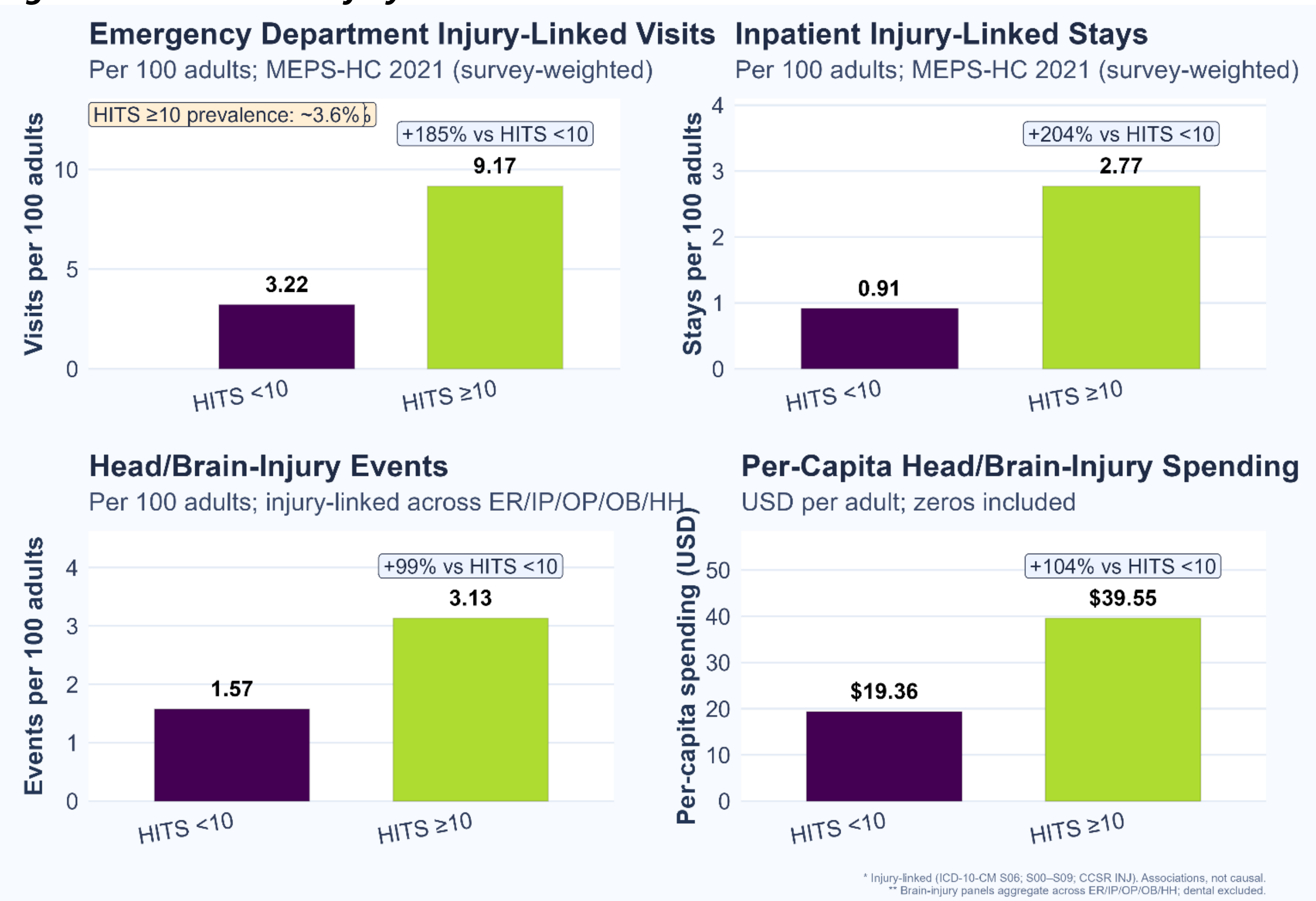
[1] Spending and events are injury-linked (diagnosis-based) and cannot establish that injuries were caused by IPV.
[2] Per-capita values are averaged across all adults in the group; individuals without head/brain injury contributions are $0.

Magnitude: Adjusted risk ratios for key adversities range from 1.87 for other/multiple race to 4.61 for feeling stressed, with housing problems at 2.88 and little/no life satisfaction at 1.51 (Table 3).

Figure 1. IPV & Brain Injury in U.S. Adults



Implications of the research

Head/brain-injury burden is substantially higher among adults who screen IPV-positive, underscoring the need to integrate brain-injury screening and follow-up into IPV response. For policymakers, this means that strategies to address IPV must explicitly consider TBI and its long-term consequences. Prevention and intervention programs must be resourced not only to reduce the incidence of violence-related harm but also to ensure that survivors receive appropriate screening, treatment, and follow-up care for head trauma.

Care is concentrated in acute settings (ER and inpatient), with markedly higher ER event rates among IPV-positive adults. Better integration of screening, documentation, and referral pathways across emergency, trauma, and behavioral health services is warranted. A key implication points to the need for integrated approaches where survivors interact, including but not limited to the criminal legal system, social services, and community-based IPV advocates, to identify and respond to survivors at the highest risk. These institutions should consider policies that require standardized screening and referral pathways so that injuries are not missed and survivors can access appropriate medical, legal, and social supports simultaneously. Specific recommendations follow.

Recommendations

- *Standardized protocols*: Implement routine protocols so that IPV-positive patients with head injury are screened for brain injury, and patients with head injury are also screened for IPV.
- *Integrated referral pathways*: Establish clear referral systems across emergency, primary, and specialty care so that when screening protocols are triggered, patients are automatically connected to appropriate follow-up services.
- *Follow-up for delayed symptoms*: Incorporate temporal follow-up in care plans, recognizing that brain injury symptoms may appear or worsen over time.
- *Accessibility and appropriateness*: Ensure screening and follow-up services are accessible, widely available, and tailored to the needs of IPV survivors.
- *Holistic screening*: Extend screening beyond identifying head injury to include social and health needs that may magnify injury effects and increase vulnerability to future IPV.
- *Care coordination*: Utilize medical social workers and interdisciplinary teams to integrate medical, psychosocial, and safety needs into a comprehensive care response.

References and further reading

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About your organization

The **I**nvestigating **S**patial **S**tructures in **U**rban **E**nvironments (ISSUE) Lab at The Ohio State University conducts cutting-edge research at the intersection of public health, law, and social work, with a focus on violence-related harm and structural inequality. Using advanced statistical and geospatial methods, we translate research into policy, practice, and prevention strategies to protect vulnerable populations and promote equity.

Contact information:

Gia Elise Barboza-Salerno, MA, MS, JD, PhD
Colleges of Social Work and Public Health
The Ohio State University
barboza-salerno.1@osu.edu

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Methodology

We analyzed the Medical Expenditure Panel Survey (Household Component; MEPS-HC) for the calendar year 2021, focusing on adults aged 18 years and older. The analysis incorporates the Person file (which provides weights, strata, and primary sampling units), the Conditions file (containing ICD-10-CM diagnoses and CCSR groupings), the Event files (including emergency room, inpatient, outpatient, office-based, home health, and prescribed medicines), and the condition–event linkage file. Since 2021, dental events cannot be reliably linked to conditions, dental care is excluded from condition-specific analyses and is, when shown, treated and labelled separately. All estimates use MEPS person-level weights (PERWT21F) and account for the complex survey design with VARSTR and VARPSU to produce nationally representative results estimates.

Key measures

Our primary exposure is screening for IPV based on the HITS scale from the MEPS social-determinants items. We classify respondents as screen-positive for IPV if HITS is 10 or higher and screen-negative if HITS is lower than 10. For descriptive purposes, we also separate scores of 5–9. Our injury identification focuses on brain injuries. We identify TBIs using ICD-10-CM code S06 and non-TBI head injuries using codes S00–S09. We categorize "other injuries" as ICD-10-CM codes outside the head categories or CCSR codes beginning with "INJ." Injuries are initially flagged in the Conditions file and then linked to specific events, allowing us to attribute emergency room, inpatient, and other care to injury, as well as brain injury when applicable. Our outcomes include injury events per 100 adults (covering emergency and inpatient encounters), per-person injury spending (including emergency, inpatient, and prescription expenditures in 2021 USD), and specific rates, spending, and weighted counts related to brain injuries. All per-capita metrics are annualized and include zeros for individuals with no recorded injury events.

Linkage and variable construction

Linking conditions to events involves matching DUPERSID and CONDIDX from the Conditions file with EVNTIDX in each Event file. This identifies which encounters are injury-related and, when applicable, which are related to brain injuries. Medications are linked to their corresponding event or condition through the linkage file, using LINKIDX, when available. For each Event file, injury-linked counts and spending are combined at the person level and then merged with the Person file using DUPERSID to create person-level totals used in analysis.

Analytic approach

First, we estimate the survey-weighted prevalence of IPV screen positivity, which is approximately 3.6% and corresponds to around 9.0 million U.S. adults in 2021. To adjust for comparisons of "per 100" rates and per-capita dollars, we fit survey-weighted generalized linear models (*svyglm*) that include age band, sex, and race/ethnicity as covariates. We model counts (events per 100 adults) using a quasi-Poisson family and log link, and dollars using a Gaussian family and identity link, which gives us the mean dollars per person. Based on these models, we calculate marginal (standardized) adjusted means for each HITS group and report absolute and percent differences relative to the HITS < 10 reference group (i.e., negative IPV-screen). For binary outcomes (such as factors associated with IPV positivity), we use log-link models to estimate adjusted prevalences and risk ratios. The standard errors and confidence intervals account for the MEPS complex design, and we denote statistical significance in tables at $p < 0.05$, $p < 0.01$, and $p < 0.001$.

Interpretation notes and limitations

Our findings show cross-sectional results for a single calendar year, expressed in 2021 dollars. Since dental events can't be reliably tied to conditions in 2021, any dental spending data presented is clearly labeled and interpreted with caution. We use ICD-10-CM and CCSR groupings to categorize injuries. TBI (S06) events are rare but often severe, especially for inpatient care, so some estimates are based on small numbers and have wider confidence intervals. Although our models account for key demographics, there may still be some residual confounding, and our results should be viewed as associations rather than causes and effects.

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Table 1. ER visits and spending patterns of adults with and without a positive IPV-screen

Outcome (per person unless noted)	Ref: IPV-screen negative	IPV-screen positive	Δ vs Ref (positive)	%Δ (positive)
ER visits (per 100 adults)	3.22	9.17	5.95	+185%
Inpatient stays (per 100 adults)	1.57	3.13	1.56	+99%
ER spending ($)	40.1	110	69.7	174
Inpatient spending ($)	191	382	191	99.9
Prescription spending ($)	23.3	17.2	−6.12	−26.3

Table 2. Demographic characteristics of adults with and without a positive IPV-screen

Variable	Level	Ref: IPV-negative	IPV-positive	RR (>=10 vs <10)
Age Category	18–24	9.5	17.5	1.855 (1.430–2.407) ***
	25–34	17.7	23.9	1.345 (1.114–1.624) **
	35–44	16.9	19	1.124 (0.915–1.380)
	45–54	15.6	13.5	0.863 (0.699–1.065)
	55–64	16.8	12.2	0.729 (0.580–0.916) **
	65–74	13.8	9.5	0.687 (0.522–0.904) **
	75–84	7.3	4.1	0.558 (0.366–0.850) **
	85+	2.4	0.4	0.151 (0.065–0.350) ***
Sex	Female	51.5	58.4	1.134 (1.043–1.233) **
	Male	48.5	41.6	0.858 (0.764–0.963) **
Race/Ethnicity	Hispanic	16.8	17.2	1.020 (0.839–1.240)
	NH White only	62.2	55.3	0.889 (0.809–0.977) *
	NH Black only	11.7	17.1	1.455 (1.128–1.878) **
	NH Asian only	6.2	4.8	0.779 (0.492–1.234)
	NH Other/Multiple	3	5.7	1.868 (1.209–2.885) **
Region	Northeast	17.4	14.5	0.836 (0.626–1.118)
	Midwest	20.8	18	0.867 (0.706–1.065)
	South	37.9	43.3	1.142 (1.015–1.285) *
	West	23.9	24.1	1.009 (0.850–1.198)

Table 3. Adjusted Prevalence by sociodemographic characteristics and health-related social needs (HRSNs)

Section	Item	Adj prevalence: HITS<10	Adj prevalence: HITS >=10	RR (>=10 vs <10)
Education	Less than high school	11.8	19	1.608 (1.346–1.922) ***
	High school or GED	26.8	28.6	1.070 (0.904–1.268)
	College degree or more	61.4	51.2	0.834 (0.753–0.924) ***
Marital status	Never married	12.4	23.7	1.910 (1.643–2.218) ***
	Married or widowed	59.6	39.3	0.659 (0.579–0.749) ***
	Divorced or separated	28	33.6	1.201 (1.101–1.311) ***
Family income (% FPL)	<100%	9.5	19.7	2.084 (1.722–2.523) ***

	100–199%	3.5	6	1.720 (1.186–2.496) **
	200–399%	11.2	16.8	1.493 (1.191–1.874) ***
	≥400%	28.1	26.8	0.951 (0.806–1.122)
Insurance (18–64)	Any private	47.7	28.1	0.589 (0.499–0.696) ***
	Public only	74.9	54.2	0.724 (0.657–0.797) ***
	Full-year uninsured	16	30.7	1.918 (1.625–2.264) ***
Health-Related Social Needs (HRSNs)	Little or no life satisfaction	9.1	13.7	1.505 (1.138–1.990) **
	Feeling stressed	5.4	25.6	4.612 (3.247–6.553) ***
	Housing problems	17.9	58.7	2.878 (2.450–3.380) ***
	Food insecurity	30.9	59.2	1.842 (1.553–2.186) ***
	Financial problems	19	55.7	2.868 (2.336–3.522) ***
	Social network problems	32.6	71.8	2.079 (1.845–2.343) ***
	Fair/poor crime safety	18.6	58.9	3.004 (2.495–3.617) ***
	Faces discrimination	13.4	29.3	2.143 (1.643–2.795) ***

Table 4. Health Conditions of adults with and without a positive IPV-screen

Section	Item	Adj prevalence: IPV-screen negative	Adj prevalence: IPV-screen positive	RR (positive vs negative)
Chronic conditions	High blood pressure	33	41	1.243 (1.100–1.404) **
	Heart disease	4.8	6.9	1.432 (1.023–2.005) *
	High cholesterol	31.4	35.7	1.137 (1.004–1.287) *
	Cancer	11.3	13.2	1.167 (0.897–1.518)
	Diabetes	11.1	14.7	1.323 (0.922–1.900)
	Asthma	14	21.4	1.524 (1.266–1.835) ***
	Stroke	3.8	5.4	1.445 (0.947–2.203)
	Emphysema	1.4	4.5	3.111 (1.862–5.196) ***
	Arthritis	25	37.8	1.511 (1.324–1.725) ***
No. of chronic conditions	0	40.4	20.2	0.523 (0.369–0.739) ***
	1	41	32.3	0.788 (0.701–0.885) ***
	2	23.7	24.6	1.035 (0.876–1.223)
	3	16.1	19.3	1.195 (0.986–1.449)
	4+	11	14.6	1.329 (1.048–1.685) *
Limitation	Any limitation	19.9	39.5	2.190 (1.775–2.704) ***
Perceived physical health	Excellent/very good	59.5	43.9	0.738 (0.661–0.824) ***
	Good	29.1	29.8	1.024 (0.886–1.184)
	Fair/poor	11.3	28.5	2.515 (2.119–2.986) ***
Mental health	PHQ-2 ≥3 (probable depression)	1.5	9.5	6.031 (3.293–11.048) ***
	Moderate (K6 7–12)	7.8	9.5	1.215 (0.702–2.102)
	Serious (K6 ≥13)	2.7	17.9	6.163 (4.249–8.939) ***
Current smoking	Current smoking	12.0	26.9	2.204 (1.654–2.936) ***

Table 5. Emergency department (ER) injury-linked encounters and spending by category (unweighted events; total dollars).

Injury category	Number of Events	Spending
Fracture	393	395,796
Injury – unspecified	298	414,069
Other injury	225	188,442
Head injury – other (non-TBI)	69	111,105
Head injury – TBI	27	42,023

Table 6. Inpatient (IP) injury-linked discharges and spending by category (unweighted events; total dollars).

Injury category	Number of Events	Spending
Injury – unspecified	96	2,068,418
Fracture	92	1,857,471
Other injury (S/T)	69	725,846
Head injury – TBI	9	175,246
Head injury – other (non-TBI)	6	122,097

Table 7. Brain-Injury Prevalence, Events, and Per-Capita Spending

	Weighted # brain injury	Per-cap $ Brain	Brain events / 100	Total $ Brain (B)	Brain injury % (within BI)	Brain injury prevalence (%)
No IPV	586,909	21.78	1.25	2.88	43.54%	0.44
IPV-screen negative	693,630	19.36	1.57	1.77	51.46%	0.64
IPV-screen positive	67,295	39.55	3.13	0.35	4.99%	0.75